

Component	Responsibilities	Interacts With
App Executive (main.py)	Controls the overall CLI flow. Displays menus, receives user inputs, and coordinates calls to modules for user registration, matching, messaging, and notes.	User Interface (CLI), User Repository, Matching Service, Messaging Service, Notes Manager, Utility, Database
User Interface (CLI)	Displays menu options, collects user input (e.g., login, search, match, message). Handles user feedback and result output.	App Executive
User Repository (user_repo.py)	Handles all CRUD operations for user data (create, read, update, delete). Fetches user profiles, manages login and registration.	App Executive, Database
Matching Service (matching.py)	Calculates match scores based on attributes (e.g., city, hobby, language, gender, age range). Suggests compatible users.	App Executive, User Repository, Database
Messaging Service (message.py)	Manages sending, viewing, and marking messages between users. Stores messages in the database with timestamps.	App Executive, Database
Notes Manager (note.py)	Handles creating and retrieving notes related to users. Stores and retrieves note data.	App Executive, Database
Utility Service (utility.py)	Validates user input (e.g., checking for blanks, ensuring numeric values). Helps maintain smooth CLI flow.	App Executive
Database Layer (database.py)	Manages SQLite database connection and schema initialization (init_db(), get_conn()). Provides a consistent access point for data operations.	All services needing database access
User Data Storage (users table)	Stores user profiles: username, age, city, hobby, gender, language.	User Repository, Matching Service
Message Data Storage (messages table)	Stores messages: sender, receiver, content, read status, timestamp.	Messaging Service
Note Data Storage (notes table)	Stores personal notes related to other users.	Notes Manager

